Reelection Incentives and Local Public Performance. With Evidence from a Municipal Reform in Mexico

Abstract

This paper studies how electoral career concerns affect local public performance. I develop a moral hazard model and compare the combination of explicit and implicit incentives with the case of bonus payments only. I find that career concerns are more effective in smaller communities with tighter competition, while bonus payments become more relevant otherwise. I test these predictions using a reform introducing reelection for municipal presidents in Mexico, and exploiting the staggered implementation across states. A difference-in-differences estimation shows that reelection increases municipal nighttime luminosity by about 0.25 log points. The effect is robust across specifications, stronger in smaller municipalities, and weaker for incumbents who won by large margins.

1 Introduction

Elections are usually treated as the main incentive device for politicians. If voters can reward incumbents with another term, then politicians may work hard even without direct performance pay. However, such mechanisms might not always work, as electoral incentives depend on how much reelection is worth, how strongly effort affects the probability of winning again, and how effective alternative monetary incentives are. When political competition is weak or the information of voters is noisy, career concerns may provide too little discipline and bonus payments can become more relevant.

This paper studies the interaction between implicit electoral incentives and explicit monetary incentives for politicians. I develop a simple model in which a risk-averse politician chooses effort, voters decide whether to reelect the incumbent, and the government chooses the compensation scheme. The model combines political career concerns with a standard multitask incentive problem. The central prediction is that reelection raises effort when future office rents are valuable relative to the value of the tasks, but the strength of this effect depends on the competitiveness and informativeness of elections. When reelection is less responsive to effort, optimal direct incentives should become stronger.

The empirical part tests the implications of the model using the introduction of municipal reelections in Mexico. In 2014, a reform allowed the consecutive reelection for all oficials entering duty after the transition. Notably, states implemented the reform at different times because municipal election calendars were not aligned within the country. I use this variation to estimate the effect of introducing reelection incentives on municipal performance, measured by monthly nighttime luminosity.

The main result is that reelection incentives increase local economic activity. In the preferred staggered difference-in-differences specification, introducing reelection raises luminosity by about 0.25 log points. The estimate is robust to term-cycle controls, politician controls, and municipality-specific trends and seasonality. Population-weighted estimates are smaller, suggesting that the effect is driven by smaller municipalities.

The results of a heterogeneity analysis of treatment effects are mostly consistent with the proposed theoretical mechanism. Effects are stronger in smaller municipalities, where the value of holding office may be large relative to measured economic output. Politicians who won by larger margins appear less responsive to the reform, consistent with the idea that reelection incentives are weaker when incumbents are electorally safer. Second-term politicians are also less affected than first-term politicians, as expected when the career concerns channel disappears. While treatment effect is positive throughout the whole term, it is largest in the middle of the cycle, rather than immediately before the next election, slightly contrary to what theory suggests.

I also examine whether the estimates are likely to be driven by selection rather than effort. The available politician-level data are limited, but reelection does not appear to generate large changes in candidate gender or winning margins. The reform is associated with a smaller increase in population than in luminosity, which is consistent with an increase in economic activity per capita rather than only a change in municipal size. These tests are not conclusive, but they support the interpretation that the reform changed incumbent efforts rather than abilities.

The policy implications of such results are quite straightforward. First, my project shows that while wages might not be as important in the setting of small communities, explicit bonus payments should not be ignored in many other cases. In particular, it might be reasonable to provide larger monetary incentives to politicians that are rather certain of getting reelected. Additionally, it might prove beneficial to vary the size of bonus payments within term cycles.

My theoretical contribution is to link two incentive channels that are often studied separately. Models of electoral accountability show how voters discipline incumbents through reelection (Barro, 1973; Ferejohn, 1986; Persson and Tabellini, 2002), while career-concerns models show how future rents can substitute for explicit pay (Gibbons and Murphy, 1992; Holmström, 1999). Recent political-agency work emphasizes that these mechanisms also depend on dynamic accountability (Ashworth et al., 2017; Acharya et al., 2025). My thesis treats electoral competition as a part of the optimal compensation problem: when implicit incentives are weak, stronger explicit incentives may be optimal.

The paper also relates to work on multitasking, politician pay, and selection. Holmström and Milgrom (1991) shows that performance pay can distort effort toward measured tasks, while Alesina and Tabellini (2007, 2008) apply related trade-offs to politicians and bureaucrats. Closest to the model, Gersbach and Liessem (2008) and Britz et al. (2022)

study incentive pay for politicians and policy-makers, and Besley (2004) analyzes politician pay and selection. Contrary to most work in this area, I focus on the heterogeneity of the two incentive mechanisms, suggesting when each of them might be more beneficial.

Empirically, my project contributes to evidence on how political incentives affect behavior. Term limits and reelection prospects change policy choices and corruption (Besley and Case, 1995; Ferraz and Finan, 2011), and electoral pressure changes legislative behavior (Fouirnaies and Hall, 2022). Elections can also distort effort toward opportunistic or visible policies (Akhmedov and Zhuravskaya, 2004; Brollo et al., 2013). The Mexican reform provides a setting in which reelection concerns change sharply, allowing a better identification strategy to study how implicit incentives affect local performance.

The remainder of the paper proceeds as follows. Section 2 presents the model and its empirical predictions. Section 3 describes the Mexican reelection reform, the data, and the identification strategy. Section 4 reports the empirical results. Section 5 concludes.

2 Theory

This section develops a simple model linking reelection incentives to politicians' effort. The model combines two standard ideas. First, as in linear incentive models, politicians may be paid according to noisy performance signals (Holmström and Milgrom, 1987, 1991). Second, as in models of political career concerns, voters use observed outcomes to decide whether to reelect incumbents (Persson and Tabellini, 2002; Dewatripont et al., 1999). The contribution of the framework is to put these forces together with a risk-averse politician: effort can be motivated either by direct monetary incentives or by the prospect of future political rents, but both channels expose the politician to risk.

The model is deliberately stylized. Its purpose is not to describe every institutional detail of Mexican municipalities, but to organize the empirical predictions. The key result is that reelection should raise effort when future rents are valuable, especially when direct performance pay is noisy or difficult to design. The model also predicts that the effect should be stronger in settings where reelection prospects are responsive to effort and where the value of future office is large relative to the social value of current output.

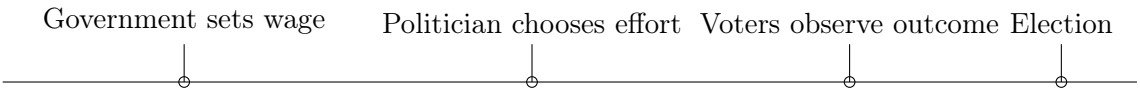
Subsection 2.1 introduces the players and timing. Subsection 2.2 solves for effort and compensation. Subsection 2.3 summarizes the empirical implications and limitations. Additional derivations can be found in Appendix A.

2.1 Setup

There are three players: a politician, voters, and a government. The politician chooses effort, voters decide whether to reelect the politician, and the government sets the politician's compensation. In the baseline model effort is one-dimensional. This keeps the

exposition transparent; Appendix A.3 shows how the logic extends to multiple tasks.

The timing is as follows. First, the government commits to a wage schedule. Second, the politician chooses effort. Third, voters observe a noisy outcome, update beliefs about the politician's ability, and decide whether to reelect or not.



2.1.1 Politician

The politician chooses effort $t \geqslant 0$ at private cost $C(t) = \frac{1}{2}ct^2$, where $c > 0$. Compensation is linear in a noisy signal $s = t + \eta$:

$$w = f + as,$$

where f is fixed pay, $a \geqslant 0$ is the piece rate, and $\eta \sim N(0,\sigma_\eta^2)$ is measurement noise. If reelected, the politician receives the discounted value of future office, δV, where $V \equiv R+g$. Here $R \geqslant 0$ is an ego-rent from holding office, $g \geqslant 0$ is an additional future payment set by the government, and $\delta \in [0,1]$ is the discount factor.

The politician has mean-variance preferences. That is, they solve

$$\max_{t \geqslant 0} u(x(t)) = \mathbb{E}[x(t)] - \frac{1}{2} r Var[x(t)], \tag{1}$$

where $r \geqslant 0$ captures risk aversion and

$$x(t) = f + a(t + \eta) - \frac{1}{2}ct^2 + \delta V \cdot \mathbb{I}\{\text{reelected}\}.$$

The mean-variance form is a tractable certainty equivalent. Appendix A.2 shows that the main qualitative results are unchanged with CARA utility.

2.1.2 Voters

Voters receive benefit

$$B(t) = v(t + \theta + \varepsilon),$$

where $v > 0$ is the social value of effort, $\theta \sim N(0,\sigma_\theta^2)$ is the politician's ability, and $\varepsilon \sim N(0,\sigma_\varepsilon^2)$ is noise in observed performance. The shocks θ, η, and ε are independent.

Voters observe $B(t)$, and, understanding the equilibrium strategy, they infer ability as

$$\tilde{\theta} = \frac{B(t)}{v} - \bar{t} = \theta + t + \varepsilon - \bar{t},$$

where $\bar{t}$ is the equilibrium effort voters expect. They reelect the incumbent if inferred ability exceeds the expected ability of challengers $\mathbb{E}[\theta] = 0$. The reelection probability is

$$P(t) = 1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t}).$$

Here $\tilde{t} = \bar{t} - t$, and Φ_{σ^2} and ϕ_{σ^2} denote the cdf and pdf of a normal distribution with variance σ^2. In equilibrium, $\tilde{t} = 0$ and $\partial\tilde{t}/\partial t = -1$. Thus, effort raises reelection probability at rate $\bar{\phi} \equiv \phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(0) = (2\pi(\sigma_\theta^2 + \sigma_\varepsilon^2))^{-\frac{1}{2}}$.

2.1.3 Government

The government is benevolent, does not collude with the politician, and can commit to pay g. It chooses f, a, and g to maximize expected voter surplus. Namely, it maximizes

$$VS = vt^* + v\theta - \mathbb{E}[w(t^*)], \tag{2}$$

subject to incentive compatibility (IC) and participation constraints (IR):

$$t^* \in \arg\max_{t \geqslant 0} u(x(t)), \qquad \mathbb{E}[u(x(t^*))] \geqslant 0.$$

The outside option is normalized to zero. This normalization permits negative fixed pay f, while a and g are restricted to be non-negative.

2.2 Solution

The model is solved by backward induction. I derive the politician's effort choice (2.2.1) and then the government's optimal compensation policy (2.2.2).

2.2.1 Optimal Effort

As a benchmark, the first-best effort maximizes $vt - \frac{1}{2}ct^2$, so $t^{FB} = v/c$. Under performance-pay only optimal compensation is $a = v/(1 + cr\sigma_\eta^2)$, inducing $t^{H-M} = v/[c(1 + cr\sigma_\eta^2)]$

In my setup, the politician maximizes the mean-variance certainty equivalent with

$$\mathbb{E}[x(t)] = f + at - \frac{1}{2}ct^2 + \delta V\left[1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right],$$

and

$$Var[x(t)] = a^2\sigma_\eta^2 + \delta^2 V^2 \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\left[1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right]. \tag{3}$$

Appendix A.1.1 derives Equation 3. The first term is the risk created by performance pay; the second is the risk created by uncertainty about reelection.

Dropping terms that do not vary with effort, the politician's problem is

$$\max_{t \geqslant 0} \left\{ at - \frac{1}{2}ct^2 + \delta V \left[1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right] - \frac{r\delta^2 V^2}{2} \left[\Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t}) - \Phi^2_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right] \right\}. \tag{4}$$

The first-order condition is

$$a + \delta V \phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t}) + \frac{r\delta^2 V^2 \phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})}{2} \left[1 - 2\Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right] = ct. \tag{5}$$

The marginal benefit of effort has three components: direct pay a, a higher probability of obtaining future rents δV, and a change in the variance of reelection returns. In the symmetric equilibrium, $\tilde{t} = 0$ and $\Phi(0) = 1/2$, so the variance term drops out. Therefore,

$$t = \frac{a + \delta V \bar{\phi}}{c}.$$

Proposition 1. *Holding compensation fixed, effort decreases with uncertainty about ability (σ_θ^2) and noise in voter information (σ_ε^2).*

When observed performance is noisy, an additional unit of effort has a smaller effect on the probability of reelection. Political career concerns therefore become weaker.

2.2.2 Optimal Compensation

The participation constraint binds through the fixed payment f. Substituting the binding participation constraint and equilibrium effort into voter surplus makes the government's problem equivalent to maximizing joint surplus:

$$SW = vt^* + v\theta - \frac{1}{2}c(t^*)^2 + \frac{\delta R}{2} - \frac{r\delta^2 V^2}{8} - \frac{1}{2}ra^2\sigma_\eta^2.$$

Using $t^* = (a + \bar{\phi}\delta V)/c$, this becomes a problem in a and g:

$$v\frac{a + \delta\bar{\phi}(R+g)}{c} - \frac{c}{2}\left(\frac{a + \delta\bar{\phi}(R+g)}{c}\right)^2 - \frac{ra^2\sigma_\eta^2}{2} - \frac{r\delta^2(R+g)^2}{8}.$$

The first-order conditions are

$$\frac{v}{c} - \frac{a + \delta\bar{\phi}(R+g)}{c} - ra\sigma_\eta^2 \leqslant 0, \tag{6}$$

$$\delta\bar{\phi}\frac{v}{c} - \delta\bar{\phi}\frac{a + \delta\bar{\phi}(R+g)}{c} - \frac{r\delta^2(R+g)}{4} \leqslant 0,$$

with equality when a,$g > 0$. Solving the system with the non-negativity constraints gives

$$\begin{cases} g = \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2} - R, & a = \frac{v}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2}, & \text{if } R \leqslant \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2}, \\ g = 0, & a = \frac{v-\delta\bar{\phi}R}{1+cr\sigma_\eta^2}, & \text{if } \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2} \leqslant R \leqslant \frac{v}{\delta\bar{\phi}}, \\ g = 0, & a = 0, & \text{if } \frac{v}{\delta\bar{\phi}} \leqslant R. \end{cases} \tag{7}$$

When intrinsic office rents R are small, the government supplements them with g because reelection incentives are valuable. When R is intermediate, the government relies on office rents but still uses direct performance pay. When R is sufficiently large, career concerns alone induce enough effort, and direct performance pay is unnecessary.

Appendix A.3 extends the model to multiple tasks. Let $B'(t)$ be the vector of marginal social benefits and let $[C_{ij}]$ be the Hessian of the cost function. The interior analogue is

$$a = (I + r[C_{ij}]\Sigma_\eta)^{-1} \left[B'(t) - \delta\bar{\phi}V\mathbf{1}\right]. \tag{8}$$

This is the standard multitask incentive formula, adjusted for the fact that reelection rents already reward all tasks that voters observe. Career concerns therefore crowd out monetary incentives most strongly for tasks that are well-perceived by voters but hard to measure institutionally.

2.3 Implications and Limitations

To connect the model to the empirical analysis, compare effort with and without reelection. Without reelection, we have $t = t^{H-M}$. With reelection, effort is

$$\begin{cases} t = \frac{1}{c} \frac{1+4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2} v, & \text{if } R \leqslant \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2}, \\ t = \frac{1}{c}\left(\frac{v}{1+cr\sigma_\eta^2} + \frac{cr\sigma_\eta^2}{1+cr\sigma_\eta^2}\delta\bar{\phi}R\right), & \text{if } \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2} \leqslant R \leqslant \frac{v}{\delta\bar{\phi}}, \\ t = \frac{1}{c}\delta\bar{\phi}R, & \text{if } \frac{v}{\delta\bar{\phi}} \leqslant R. \end{cases}$$

The relative gain, $\Delta t = (t - t^{H-M})/t^{HM}$, is therefore

$$\begin{cases} \Delta t = \frac{4\bar{\phi}^2\sigma_\eta^2 cr\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2}, & \text{if } R \leqslant \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2}, \\ \Delta t = \delta\bar{\phi}cr\sigma_\eta^2 \frac{R}{v}, & \text{if } \frac{v}{\delta\bar{\phi}} \frac{4\bar{\phi}^2\sigma_\eta^2}{1+4\bar{\phi}^2\sigma_\eta^2+cr\sigma_\eta^2} \leqslant R \leqslant \frac{v}{\delta\bar{\phi}}, \\ \Delta t = \delta\bar{\phi}\left(1 + cr\sigma_\eta^2\right)\frac{R}{v} - 1, & \text{if } \frac{v}{\delta\bar{\phi}} \leqslant R. \end{cases}$$

The model yields several empirical predictions.

Proposition 2. *Reelection has a larger proportional effect when the value of future office is high relative to the current social return to effort. If v rises with municipal size faster than office rents do, the effect should be stronger in smaller municipalities.*

Proposition 3. *Reelection incentives are more valuable when direct performance pay is harder to use, either because performance signals are noisy (σ_η^2 is high) or because politicians are more risk averse (r is high).*

Proposition 4. *Reelection incentives are stronger when effort has a larger effect on reelection probability. They therefore weaken as ability uncertainty (σ_θ^2) or voter-information noise (σ_ε^2) increases. They are also stronger when politicians value the future more (δ).*

Proposition 5. *In a multitask environment, reelection incentives substitute most strongly for monetary incentives on tasks that voters observe better relative to the government.*

Finally, based on derivations in Appendix A.4,

Proposition 6. *Reelection incentives are weaker when political competition is weak.*

I aim to test these propositions empirically. The main prediction is that introducing reelection should increase observed performance. The effect may be stronger in smaller municipalities if the value of office grows slower with population than the social value of tasks. It should be stronger for politicians that feel the pressure of their competition, and they should be largest late in the term if voters place more weight on recent outcomes.

There are several limitations to the model. First, the government is assumed benevolent and able to commit to verifiable signals. In reality, local politicians may influence their own compensation or manipulate measured performance. Second, the model is static except for the reelection payoff. A richer dynamic model would allow voters and politicians to learn after the first term, which could change incentives for second-term incumbents. Third, challenger and incumbent ability distributions may differ. Appendix A.4 shows that such distortions can change the strength of career concerns and may require higher monetary incentives in equilibrium.

3 Institutional Setting and Data

This section describes the institutional setting and the data used in the empirical analysis. The key institutional change is Mexico's 2014 political reform, which introduced the possibility of consecutive reelection for municipal presidents. Because states implemented the reform at different times, the setting provides useful variation for estimating how reelection incentives affect local performance.

Subsection 3.1 describes the reform in more detail, Subsection 3.2 describes the data collection process, and Subsection 3.3 presents descriptive statistics.

3.1 Setting

Mexico has a detailed federal structure, with elected officials even at the municipal level, and transparency laws requiring the publication of remuneration and salary data. In 2014,

the federal constitution of Mexico was amended, introducing the possibility of reelection for municipal presidents.

The United Mexican States have two levels of federal structure. There are 32 states (Estados), including one federal entity, Mexico City (CDMX). Each state has its own constitution, which usually restates the federal analogue and may include additional articles, as long as they do not contradict federal law. States are divided into municipalities (Municipios), with the number of municipalities varying by state. For example, Baja California Sur is divided into seven municipalities, while Oaxaca is divided into 570 municipalities. As of 2026, there are 2,457 municipalities.

Municipalities are governed by city councils (Ayuntamientos) consisting of council members (regidores and sindicos) and a president (presidente municipal). Both members and presidents of the council are elected directly by voters.

I consider a constitutional change that introduced the possibility of reelection for local politicians. Specifically, until 2014, reelection was prohibited throughout the entire country. This rule was changed with a constitutional amendment, allowing presidents and council members to be reelected for one additional *consecutive* term, as long as the term lasted no more than three years and they ran with the same party.

Technically, the amendment took effect the day after publication (10.02.2014). However, it was specified that the change would only affect those politicians who took office *after* the transition. Hence, suggested career concerns mechanisms effectively changed only after the next municipal election. Notably, while key dates vary across states, municipalities within one state are always affected simultaneously.

Figure 1: Timeline for Campeche

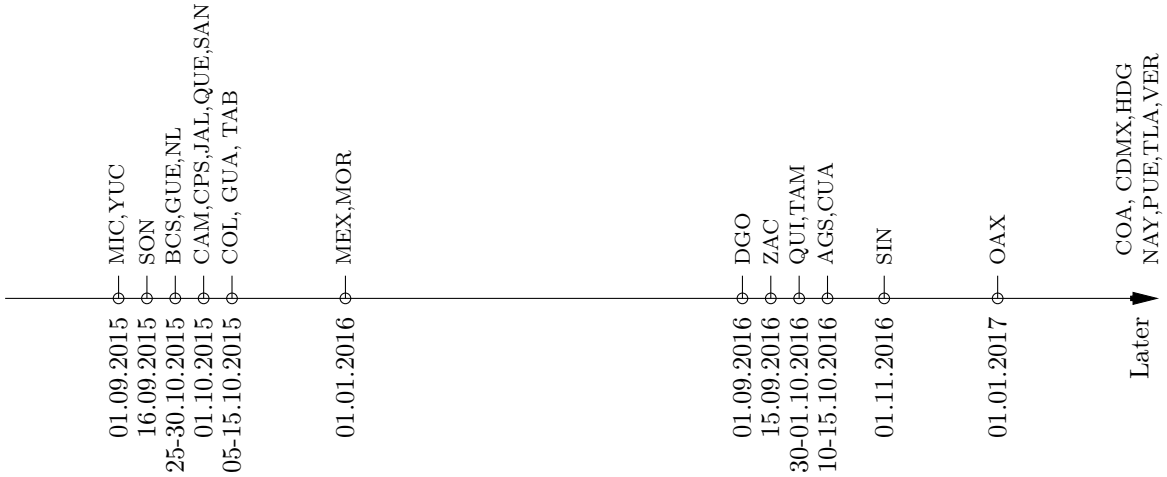


Figure 1 presents a timeline of the key dates for this amendment in the case of municipalities in the state of Campeche, which followed the most common pattern across regions. A few months after the federal amendment, Campeche also amended the state constitution. One year later, elections were held, and three months after that, the first presidents who had the option to run for a second time in 2018 began their terms. The effective treatment date would be October 1, 2015, in this case.

Fifteen other states followed a very similar timeline with treatment dates between September 1, 2015 and January 1, 2016. Nine states held the first election under the new rules in 2016, one did so in 2017, two in 2018, and two in 2021. Figure 2 shows the distribution of effective treatment dates across states and municipalities. Notably, while elections are mostly held in July, all effective changes (term starts) occur in the

last third of the year. Appendix B shows histograms with the number of transitioning municipalities (rather than states) and population-weighted histograms of transitions.

Figure 2: Effective Start Date across States



There are six special cases of transition. First, CDMX officially became a state only in 2016 and introduced reelection in 2018. Second, Puebla switched from five-year terms to three-year terms to allow for reelection, and only did so in 2018. Late transitions also occurred in Tlaxcala and Nayarit in 2021, which had transitory terms between 2016 and 2021. Hidalgo transitioned only in 2024, simultaneously reducing term length from four years. Finally, Veracruz never introduced reelection at all and continued with the four-year term format. These cases may be institutionally different, so in some specifications I run robustness checks to ensure that the effects are not driven by a particular outlier.

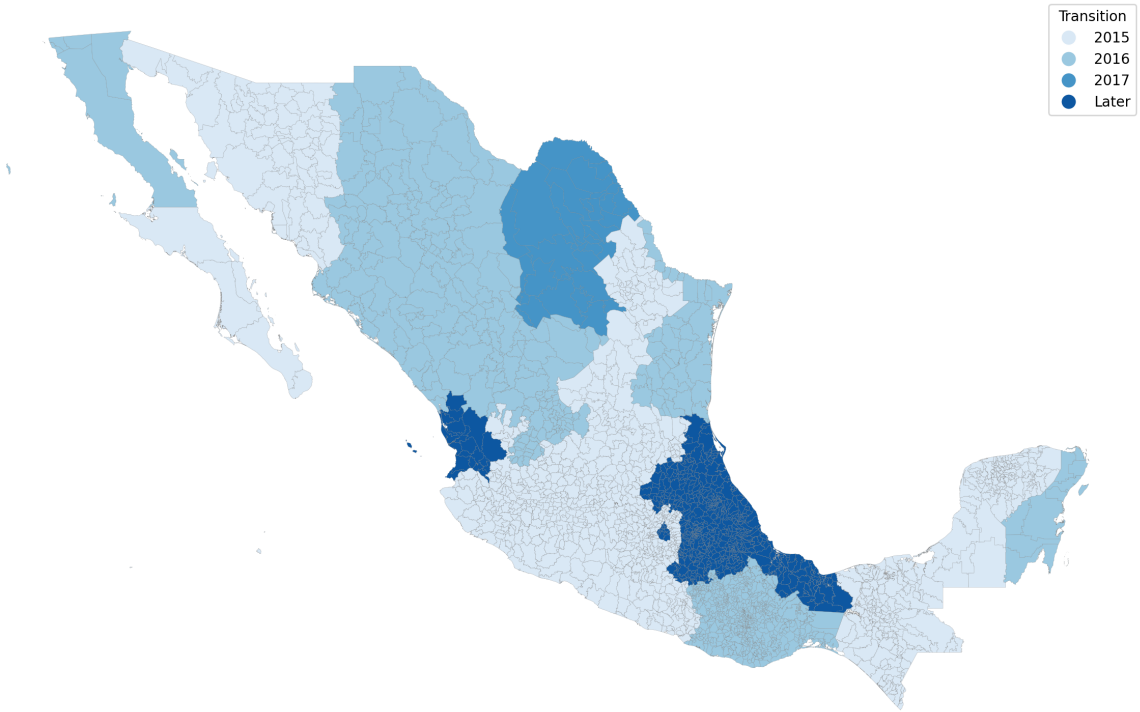
Before reelection was introduced, most municipal presidents tended to move upward (i.e. to state congress, federal congress, or governor) rather than horizontally (i.e. to a different municipality). Technically, election into neighboring municipalities was never prohibited by either the federal constitution or any state constitution. However, in practice, residency requirements often applied (Ser vecino del municipio con residencia efectiva [...] años). Hence, the possibility of reelection into a different municipality should not be a major concern for the estimation strategy (Langston, 2010).

3.2 Data Collection

I use data from several sources. First, I collect information on transition dates directly from state constitutions. Second, I take data on election outcomes from Magar (2025). Third, I download data on luminosity from Elvidge et al. (2017) as the main outcome.

To find information on transition dates for the introduction of reelection, I consider each state constitution and find the paragraph that introduces reelection. Usually, these paragraphs are direct copies of the federal version, but sometimes they have slight changes

Figure 3: Transition to Reelections in Mexican States



Note: This figure shows the first year in which incumbents with the possibility of reelection came to power in each municipality. Data are collected directly from constitutional amendments.

in wording. For each case, I find the transitory article (articulo transitorio) and the date on which it takes effect (which is always earlier than the next election).

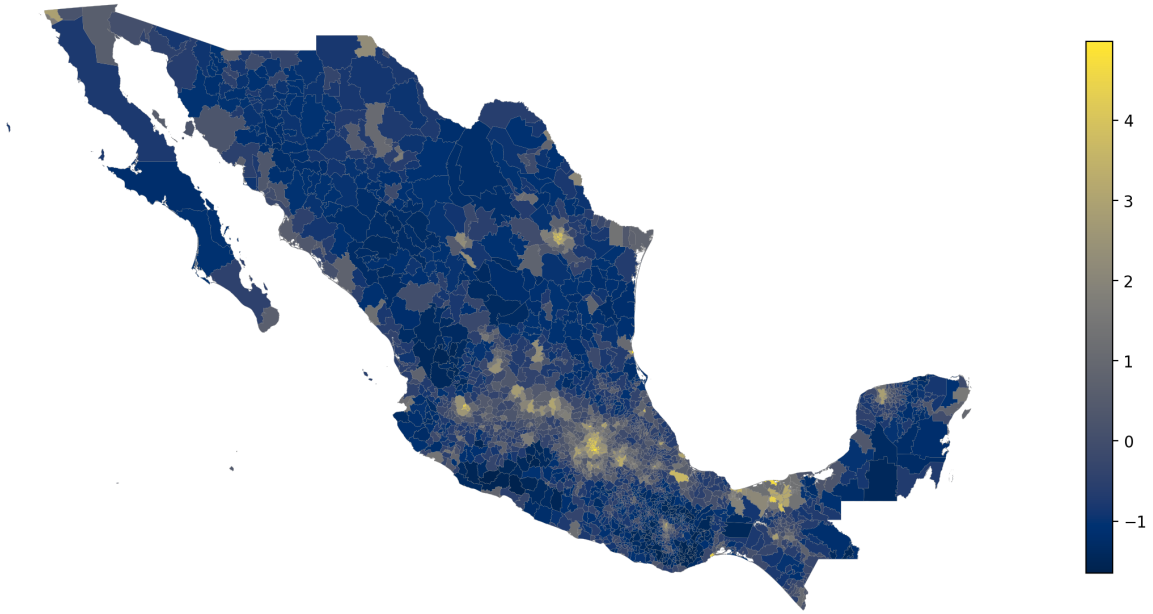
I take data on elections from Magar (2025). It contains information on the name and sex of each president elected between 1979 and 2025, as well as their party. In most cases, it also provides data on the runner-up and the winning margin in each election. Additionally, it reports post-election information about the incumbent, i.e. whether they were term-limited, whether they ran for reelection, and whether they or their party won.

Nighttime lighting has become one of the most widely used proxies for local GDP in economics (Chen and Nordhaus, 2011; Bruederle and Hodler, 2018; Pinkovskiy and Sala-i Martin, 2014) and is also suitable for this setting. I take monthly lighting data from Elvidge et al. (2017), who collect it directly from satellites and clean it for weather conditions and cloud coverage. I aggregate data from the pixel level to Mexican municipal boundaries for the months between April 2012 and January 2026.

3.3 Descriptive Statistics

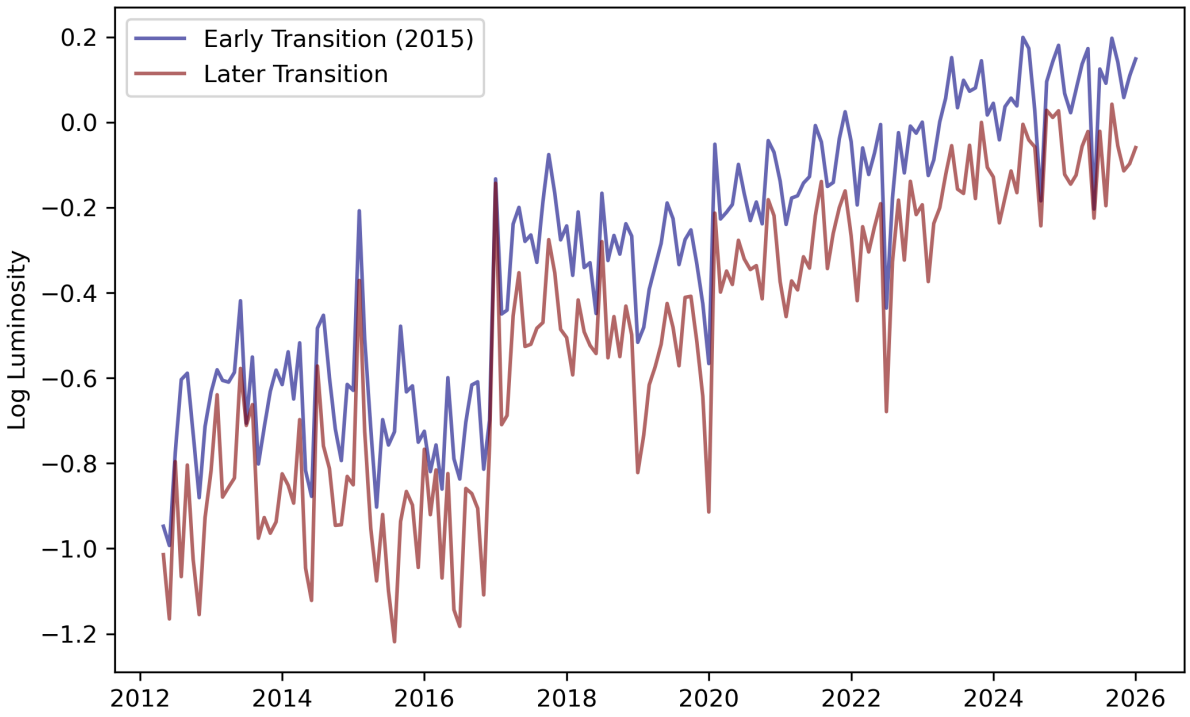
The Mexican Constitution introduced reelection on February 10, 2014. Article 115 stated that 'constitutions of states shall establish the consecutive election for the same office of municipal presidents, councilors, and trustees for an additional period, provided that the

Figure 4: Luminosity in Mexican Municipalities in 2026



This figure shows log average nighttime lighting in Mexico in January 2026. Raw data come from Elvidge et al. (2017) and are aggregated at the municipality level.

Figure 5: Luminosity Growth Rate



Mean Y: -0.482, SD: 1.344, Min: -14.22, Max: 6.40, 1%: -3.647, 99%: 3.542, N: 403,111

Note: This figure compares the growth rates of the main outcome variable – namely, log mean nighttime luminosity in a given municipality in a given year, between groups with an early transition and a late transition. Data are taken from Elvidge et al. (2017).

Table 1: Descriptives (Controls)

Variable	Control	Treated	p-Value	Number of Obs.
Log Luminosity	−0.662	−0.361	0.000	403.111
	(1.499)	(1.215)		
Log Population	9.523	9.443	0.000	402.384
	(1.501)	(1.660)		
Female	0.094	0.250	0.000	292.121
	0.292	0.433		
Win Margin	0.115	0.130	0.000	291.492
	(0.116)	(0.131)		
Second Term	0.000	0.123	0.000	407.862
	(0.000)	(0.329)		
Cartel Presence	0.106	0.133	0.000	407.696
	(0.307)	(0.339)		
Log Wages	11.120	11.115	0.259	37.350
	(0.412)	(0.427)		
Any Fixed Supplements	0.155	0.166	0.003	41.002
	(0.362)	(0.372)		
Any Bonus Payments	0.348	0.342	0.241	41.002
	(0.476)	0.474		
Share of Boni	0.044	0.044	0.902	41.002
	(0.091)	(0.097)		

This table shows the descriptive statistics for the control variables.

term of office of the city councils is not longer than three years'. Transitory Article 14 required that the new amendment did not apply to members of city councils already in office before the decree entered into force.

This amendment stands out on several dimensions. First, states implemented the law with a time lag, coming into effect only at the next election. This makes it possible to distinguish between states that had municipal elections in 2015 and those that had them in 2016 or later, allowing for a staggered difference-in-differences estimation. Second, reelection was only allowed for presidents and council members who served terms of no longer than three years. Some states had transition periods with longer terms, and the states of Veracruz and Hidalgo actually retained the four-year, no-reelection format.

Figure 3 shows the transition years for each municipality. In particular, it presents the year of inauguration of the first affected president. The figure shows some variation, with most states introducing the change between 2015 and 2017. It also appears that there is no spatial correlation or clustering of municipalities with earlier transition years.

The main outcome variable in most specifications is the logarithm of mean luminosity in each municipality. Figure 4 shows log mean lighting in Mexican municipalities in January 2026. The data align well with the spatial distribution of GDP and population:

large cities are clearly visible on the map.

Figure 5 plots the deseasonized mean log luminosity in municipalities treated in 2015 and later. The average log luminosity over April 2012 – January 2026 is about -0.5, with a standard deviation of 1.34, a minimum of -14, and a maximum of 6.5. Early and later treated units appear to have quite similar dynamics at the baseline, with almost identical growth rates over time (see Appendix B). The distribution contains heavy outliers: the first and 99th percentiles are much closer to the mean than the minimum and maximum. In most specifications, I truncate the sample at these values. The estimates in robustness checks with the whole sample are typically even larger (see Appendix C).

Table 1 shows descriptive statistics for the outcomes, the control and heterotest variables. We can see a difference between log luminosity already across means. There are more women in the treated sample, while most other variables are somewhat similar across the two groups. The difference in means for both luminosity and gender can likely explained by simple upward trends aside from the treatment effect.

4 Results

This section presents the empirical results. Subsection 4.1 presents baseline estimates of the effect of reelection on luminosity. Subsection 4.2 investigates heterogeneity, while Subsection 4.3 considers selection as an alternative mechanism and Subsection 4.4 discusses heterogeneity with respect to wages in particular.

4.1 Reelection and Luminosity

Theory suggests that politicians with the possibility of running for a second term may exert more effort, leading to better local economic outcomes. I test this prediction below.

The perfect experiment would be to randomize the introduction of reelection concerns across municipalities. In such a setting, one would randomly divide the population into two groups of municipalities that would have perfectly aligned terms, that is coinciding start and end dates, as well as simultaneous elections. In one group, we would allow politicians to be reelected for a second term, while in the other group, this would be prohibited. To focus the experiment on political effort (rather than selection), we would also fix the pool of candidates for presidents in both groups of municipalities. We could then compare the outcomes in the two groups.

One could also calculate dynamic effects and run heterogeneity tests to compare performance in different months of the term. Additionally, we could compare politicians in their first terms to politicians in their second terms (for those municipalities that are treated), large municipalities to small ones, or industrial to agricultural municipalities. Furthermore, we could randomize on other dimensions, such as term limits (number),

term length (in years), or compensation schemes, which can be relevant for further work.

Variation in the timing of amendment adoption allows the empirical design to approximate this ideal experiment by making it possible to address some (but not all) potential biases. Most of the variation originates from predetermined election calendars. To the extent that these calendars are exogenous, the reform resembles a natural experiment. Notably, the exogeneity of treatment timing is more questionable for states that decided to introduce transition terms or even chose not to switch at all.

Following Borusyak et al. (2024), I use a staggered difference-in-differences design to identify the effects of the reelection introduction. Namely, I estimate the following model:

$$Y_{mt} = A'_{mt}\lambda_m + X_{mt}\beta + D_{mt}\tau_{mt} + \varepsilon_{mt} \tag{9}$$

where Y_{mt} is the log luminosity in municipality m in month t, $A'_{mt}\lambda_m$ covers municipality-specific fixed effects and trends, $X_{mt}\beta$ specifies time-varying controls (including time fixed effects), D_{mt} is the treatment variable indicating whether reelection was introduced in municipality m by time t, and τ_{mt} is the instance-specific treatment effect, which can later be used to calculate parameters of interest (e.g., ATT).

Until recently, regressions similar to 9 were often estimated by two-way fixed effects models (e.g., Guimaraes and Portugal (2010)). However, such models impose strong assumptions on treatment effect homogeneity, and relaxing these assumptions can make the resulting estimators non-convex combinations of individual effects (De Chaisemartin and d'Haultfoeuille (2020)). I therefore use the method proposed in Borusyak et al. (2024), which, in this case, is equivalent to estimating λ_m and β on the set of untreated observations and imputing τ_{mt} as the residual. While this estimator is efficient under its assumptions and computationally fast, I complement it with other methods proposed in Callaway and Sant'Anna (2021) and De Chaisemartin and d'Haultfoeuille (2024) for robustness. A review of heterogeneous effects in difference-in-differences designs can be found in De Chaisemartin and d'Haultfoeuille (2023).

This specification allows me to compare similar treated and untreated municipalities. Most of the comparison concentrates in 2016, with 16 states treated in late 2015 and 16 states untreated at least until September 2016. The municipality and time fixed effects account for time-invariant characteristics of municipalities that adopt amendments earlier. Similarly, it accounts for municipality-invariant effects of time.

Importantly, the SDiD specification does *not* fully eliminate omitted variable biases: within-municipality time-varying shocks correlated with both reelection concerns and municipal growth rates could still distort the results. Clearly, one such confounding covariate are term cycles – fixed effects with respect to time left until the end of the term. Alongside the transition to reelectability, there is a simultaneous switch from politicians in their last year of service to politicians in the first year. Hence, I control for that in most regressions.

Table 2: Reelection and Luminosity (TWFE)

	Outcome Variable: Log Luminosity				
	(1) OLS	(2) OLS	(3) TWFE	(4) TWFE	(5) TWFE
Reelection Introduced	0.246	0.301	0.153	0.070	0.077
	(0.004)	(0.004)	(0.009)	(0.009)	(0.007)
Second Term				−0.001	0.005
				(0.005)	(0.003)
Term Length				0.034	−0.005
				(0.006)	(0.003)
Female				0.022	0.009
				(0.006)	(0.004)
Win Margin				0.097	−0.000
				(0.024)	(0.012)
Constant	−0.632	−0.591	−0.577	−0.549	1.906
	(0.003)	(0.024)	(0.006)	(0.022)	(0.031)
Cycle Controls	No	Yes	Yes	Yes	Yes
Party FE	No	No	No	Yes	Yes
Municipality FE	No	No	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes
Mun Trend & Season	No	No	No	No	Yes
Errors	Robust	Robust	Mun Cl	Mun Cl	Mun Cl
Observations	395,047	395,047	395,046	273,945	273,945
Mean of Dep. Variable	−0.485	−0.485	−0.485	−0.381	−0.381
Adj. R-squared	0.010	0.027	0.931	0.938	0.954

The outcome variable is log mean nighttime luminosity in municipality m and month t. The main regressor is a dummy equal to 1 if reelection has been introduced in municipality m before month t. Columns report OLS (1,2), and TWFE (3–5) results. Cycle controls (2–5) and politician controls (4–5) are described in the text. Standard errors are heteroskedasticity-robust (all) and clustered at the municipality level (3–5). Outcomes are trimmed at 1% and 99%.

I use the logarithm of the mean monthly nighttime luminosity as the outcome variable. I interpret it as a proxy for effort, although it may also reflect ability or other characteristics of the municipality and politician. In the main specification, I consider treatment as reelection introduced by the specified time, while it may be reasonable to distinguish between treated politicians in their first and second terms.

To control for within-term cycles (see e.g. Akhmedov and Zhuravskaya (2004)), I compute months-left-to-next-election fixed effects. Specifically, for each municipality and period I calculate how many months are left until the next election. For example, if elections occur on June 6, 2021, and inauguration takes place on October 1, the variable takes the value 1 for May, 0 for June, -1 for July, and 32 for October of that year. In most regressions, I include fixed effects for each value taken by the variable. In some cases, I

use trends or fixed effects of larger groups due to computational limitations.

As for other controls, I include the gender of the politician, party fixed effects, the win margin, and, where possible, the number of the term served or the term length. The first three variables are proxies for selection effects. The last two regressors are important as they can change contemporaneously with treatment. There were no politicians serving a second term before reelection was allowed, and these politicians may differ from the general sample. Term length may also affect total effort or its distribution across months. I therefore include these controls where possible, but they cannot be included in the SDiD specifications without modeling multiple treatments.

Table 2 reports the baseline regressions. OLS estimates are large at 0.25 log points relative to the mean of -0.5. All TWFE estimates suggest a smaller but still positive effect of the introduction of reelection. Term cycle controls increase the treatment coefficient, but neither term length nor term number appears to affect luminosity. Politician- and election-specific controls appear to reduce the effects by almost half. Municipalities with female presidents are associated with relatively higher outcomes, while closer elections (smaller winning margin) are correlated with lower performance.

The absence of any effect of being in the second term calls for further investigation. Reelected politicians may differ on two dimensions. On the one hand, they may learn by doing, so their ability may increase during the second term. On the other hand, reelection is not possible for a third time, so these presidents may have less motivation to exert effort. In Subsection 4.2, I examine this further.

Table 3: Reelection and Luminosity (DiD)

	Outcome Variable: Log Luminosity			
	(1) SDiD	(2) SDiD	(3) SDiD	(4) P-W SDiD
Treatment	0.274	0.249	0.277	0.107
	(0.015)	(0.020)	(0.037)	(0.056)
Term Controls	Yes	Yes	Yes	Yes
Politician Controls	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Mun Trend & Season	No	No	Yes	Yes
Observations	395,047	244,643	244,098	239,687
Mean of Dep. Variable	−0.485	−0.425	−0.427	−0.451

The outcome variable is log mean nighttime luminosity in municipality m and month t, trimmed at 1% and 99%. Treatment is the introduction of reelection in municipality m before month t. Term and politician controls are as in Table 2. Standard errors are clustered at the municipality level.

Table 3 shows staggered difference-in-differences estimates. The treatment appears to

increase luminosity by 0.28 log points, which is economically large. These estimates are higher than TWFE and fairly close to OLS. These results are robust to various controls, including unit-specific trends and seasonality. The two-way specification in Table 2 may underestimate the effect because of negative weighting, a problem likely to arise in long panels when later periods have relatively few controls (Jakiela, 2021).

Population-weighted estimates suggest far smaller and only marginally significant effects, which indicates that the results are driven by smaller municipalities. This aligns well with the interpretation of reelection concerns, as for most larger municipalities, these were present already before the reform through elections to state or even federal councils. I address the effect of unit size further in Subsection 4.2.

In all specifications, errors are clustered at the municipality level, while treatment appears simultaneously at the state level. This may raise some concerns regarding within-region error correlations. To address the latter, I conduct a randomization-inference test of the regression in column (2) of Table 3 with 1000 random reassignments of treatment at the state level. The p-value of the coefficient is 0.27. The distribution of the placebo effects in the randomized estimations can be found in Appendix B.

4.2 Heterogeneity Analysis

The imputation method in Borusyak et al. (2024) allows me to estimate unit-time-specific treatment effects. This is useful for analyzing the heterogeneity of treatment with respect to both time and municipality- or politician-specific characteristics. The first natural exercise is to investigate the dynamics.

Figure 6 shows the dynamic effects of the transition. The increase in luminosity is relatively small during the first year after reelection is introduced and then gradually increases. There are two visible drops after the 24th and 36th month, which I attribute to term-cycle effects. A potential concern is that the pre-trends are volatile and may even resemble a declining pre-trend.

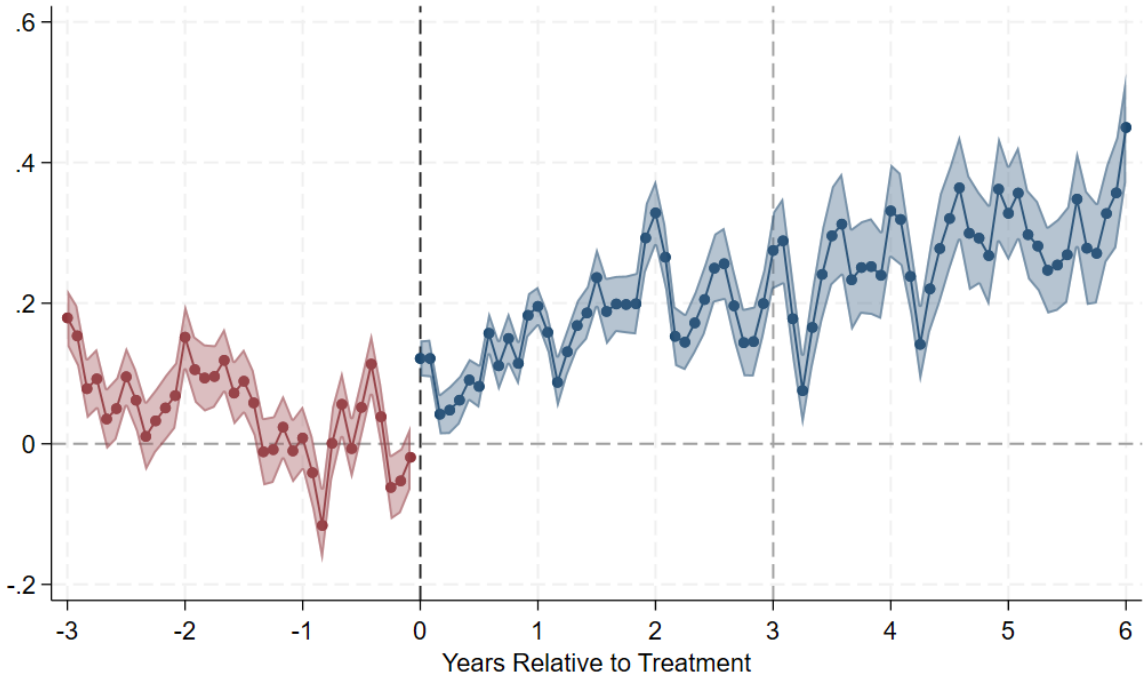
To address this issue, I turn to alternative specifications. In particular, I use a method similar to generalized synthetic difference-in-differences approaches Arkhangelsky et al. (2021). I implement the counterfactual estimator framework of Xu (2017), which extends the standard difference-in-differences approach by allowing untreated potential outcomes to follow a flexible latent factor structure rather than relying on the strict parallel trends assumption. A review of this topic and a detailed description of the particular estimator that I use can be found in Liu et al. (2024).

In the interactive fixed effects specification, untreated outcomes are modeled as

$$Y_{mt}(0) = \alpha_m + \delta_t + \lambda_m' F_t + X_{it}'\beta + \varepsilon_{mt} \tag{10}$$

where $Y_{mt}(0)$ denotes the untreated potential outcome for municipality m at time t, α_m

Figure 6: Dynamic Effects



Note: This figure shows the dynamic effects of treatment. The specification is from column (2) of Table ?? with 72 horizons (navy) and 36 pretrends (maroon).

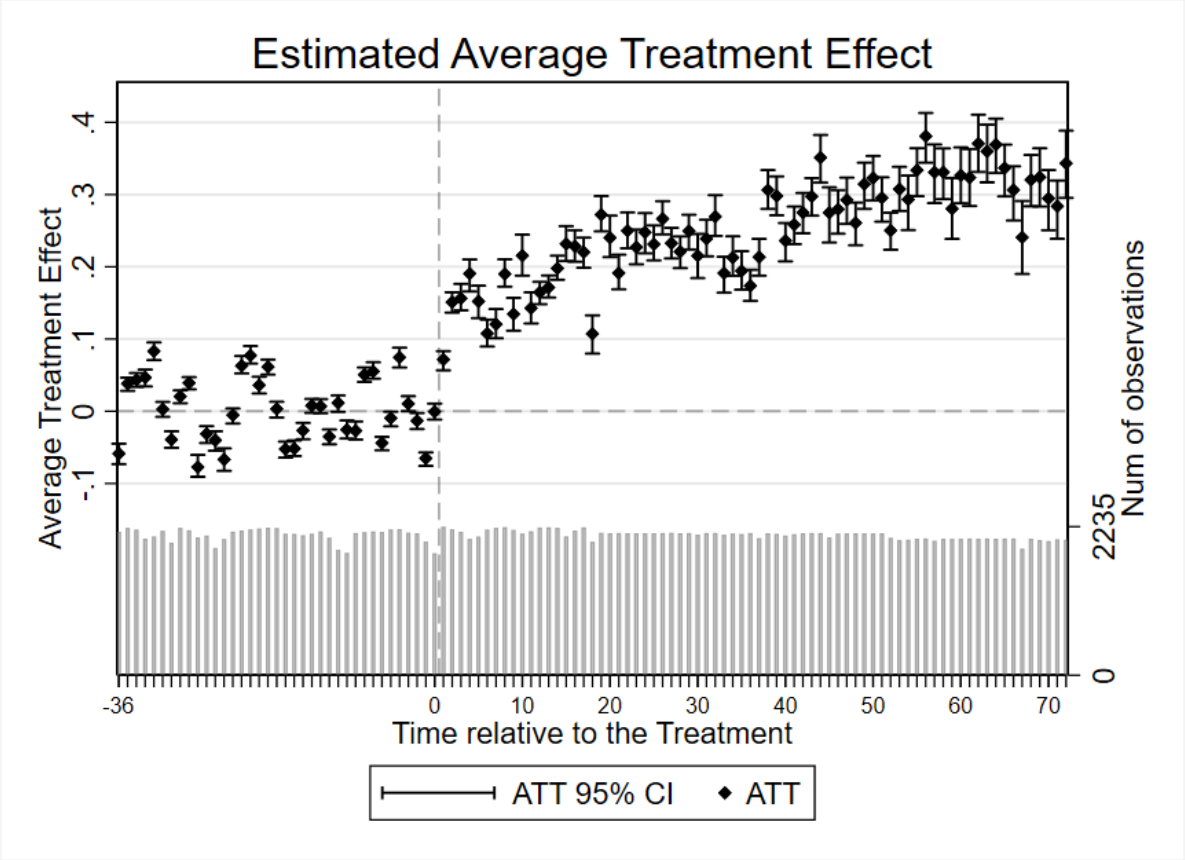
and δ_t represent unit and time fixed effects, X_{mt} is a vector of observed covariates, and λ_m captures unobserved common factors with heterogeneous unit-specific loadings. Unlike conventional two-way fixed effects estimators as in 9, this specification permits units to exhibit different underlying trends driven by latent macroeconomic, political, or regional shocks. Treatment effects are then identified by comparing observed post-treatment outcomes to the counterfactual outcomes predicted by the estimated factor structure.

Figure 7 shows the results of the interactive fixed effects estimation. Although there is significant variation in pre-treatment effects, the pre-trend disappears, and all treatment effects are positive, with an apparent upward trend and some term cyclicality. The effects are similar to the baseline specification.

Notably, term cycles are not aligned over municipalities relative to treatment timing. For municipalities treated in September, the next elections occur in 33 months, while for municipalities treated in January, this number is 29. Moreover, there is a set of states that introduce a transitory term that lasts only two years. Hence, it is reasonable to analyze dynamic effects within term cycles. In particular, I can regress the individual treatment effects on within-term cycle dummies, while controlling for baseline dynamic effects.

Table 4 shows the results. After accounting for dynamics, unit and time fixed effects, the treatment effect appears to follow a hump-shaped pattern over the term. The effect of reelection grows over the first year in office, is largest in the middle of the cycle, and

Figure 7: Dynamics with Interactive FEs



This figure shows the dynamic effects of treatment under interactive fixed effects (equation 10).

drops to its lowest point after the next election but before a new politician is inaugurated.

This pattern differs from much of the literature on electoral cycles, where expenditure is often concentrated closer to the election date. One explanation is that I analyze how term cycles *change* after treatment, rather than whether cycles exist at all. Distortive cycle effects may have been present even before the reform if politicians cared about their party or about careers outside the municipal council. The possibility of personal reelection may instead shift additional effort toward the middle of the term. In any case, this pattern warrants further investigation.

Table 4: Term Cycles

Months to Election	M>25	M24–19	M18–13	M12–7	M6–0	Post
Treatment Effect	0.142	0.173	0.204	0.188	0.127	0.100
	(0.009)	(0.014)	(0.011)	(0.007)	(0.008)	(0.006)

The table reports TWFE regression results. The outcome variable is the treatment effect from column (2) in Table **??**. The regressors are dummies for the number of months remaining until the next election. In addition to unit and time FEs, the regression controls for months-since-treatment fixed effects. Standard errors are clustered at the municipality level.

I also examine heterogeneity across municipality characteristics, such as size, and politician characteristics, gender. Winning margins may also matter: incumbents who won by large margins may expect to win again easily and may therefore respond less to reelection incentives than incumbents who faced close competition. Politicians in their second term, who cannot run again, are also likely to be less responsive than first-term politicians who gained the possibility of reelection after the reform. A simple heterogeneity analysis based on the imputation estimator is reported in Table 13 in Appendix C.

Table 5: Treatment Effect Projections

	Outcome Variable: Treatment Effect			
	(1)	(2)	(3)	(4)
Party Control	OLS	TWFE	TWFE	SS-TWFE
Female	0.005	−0.009	−0.011	−0.030
	(0.003)	(0.006)	(0.006)	(0.023)
Win Margin	0.096	−0.046	−0.050	−0.065
	(0.012)	(0.021)	(0.021)	(0.033)
Cartel Presence	0.020			
	(0.004)			
Log Population	−0.071			
	(0.001)			
Second Term			−0.004	−0.014
			(0.005)	(0.007)
Party Control	Yes	Yes	Yes	Yes
Municipality FE	No	Yes	Yes	Yes
Time FE	No	Yes	Yes	Yes
Dyn Treatment FE	No	No	Yes	Yes
Mean of Dep. Var.	0.249	0.249	0.249	0.249
Observations	157,773	157,868	154,780	54,891

The outcome variable is the treatment effect from column (2) in Table **??**. The regressors are described in more detail in the text. In columns 3,4, the regression controls for months-since-treatment fixed effects. Columns 1–3 report results on the whole sample, while column 4 reports results only for presidents who were reelected. Standard errors are clustered at the municipality level (columns 2–4).

Table 5 provides the results of regressing the treatment effect on various correlates. As predicted in the theory, smaller municipalities appear to be significantly more affected by the reform. This may be explained by the absence or limited presence of formal performance payments in smaller councils (e.g. a head of a community of 500 people), which makes implicit incentives the only mechanism with the potential to motivate effort. However, this could also be explained by the presence of reelection concerns (e.g. in state councils) after serving as non-reelectable municipal president in a larger municipal council.

Winning by a smaller margin also appears to enhance the motivating effect of the

reelection possibility. Winning with a 10% larger margin decreases the treatment effect by 0.5 log points (2% relative to the mean). This is consistent with the competition effect: it is not enough to be eligible for reelection; the incumbent must also compete for office. Finally, politicians in their second term are less affected by the reform relative to their first term (a coefficient of -0.14 log points). The fact that the effect does not return to zero may be explained by learning on the job or selection effects.

4.3 Reelection and Other Outcomes

While theory suggests increases in effort as a mechanism, selection effects could alternatively dominate. In particular, if the possibility of reelection attracted more able candidates to the pool of potential presidents, public expenditure might be spent more efficiently, simply because of that. Since I have limited politician-level data, I cannot compare education or experience levels of candidate pools before and after treatment. I therefore inspect winning margins and the gender of politicians. If selection changes substantially, electoral competition should become tougher and winning margins should decrease.

Table 6: Reelection and Other Outcomes

	Female		Win Margin		Log Population	
	(1) OLS	(2) DiD	(3) OLS	(4) DiD	(5) TWFE	(6) DiD
Reelection Introduced	0.068	0.038	−0.008	0.016	−0.001	0.017
	(0.015)	(0.031)	(0.004)	(0.010)	(0.002)	(0.003)
Female			−0.013	−0.031	−0.003	
			(0.004)	(0.011)	(0.002)	
Win Margin					−0.012	0.002
					(0.009)	(0.003)
Party FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	No	Yes	No	Yes	Yes	Yes
Time	Trend	FE	Trend	FE	FE	FE
Errors	Robust	Mun Cl	Robust	Mun Cl	Mun Cl	Mun Cl
Observations	7,763	6,474	7,763	6,474	23,198	21,407
Mean of Dep. Variable	0.184	0.164	0.125	0.121	9.867	9.863
Adj. R-squared	0.056		0.010		0.998	

The main regressor is a dummy equal to 1 if reelection has been introduced in municipality m before month t. Outcome variables and controls are described in more detail in the text. Standard errors are heteroskedasticity-robust (all) and clustered at the municipality level (cols 2,4,6).

Table 6 shows estimates of reelection effects on winning margins, the probability of a politician to be a woman, and on log population. The first two outcomes could represent proxies of selection. Difference-in-differences results are mostly insignificant, suggesting

marginal positive changes in both outcomes. This may indicate quantitatively small effects of reelection on selection, strengthening support for the effort mechanism. However, as pointed out above, these results should be considered with caution. For example, it could also be the case that the whole pool of candidates shifted to the right symmetrically, so winning margins remained constant. In any case, it is reasonable to control for both these variables as well as party fixed effects, which I do in most specifications.

Columns 5 and 6 of Table 6 report estimates of log population changes rather than luminosity. The specification is the same as column 2 in Table 3, with the outcome variable changed. There also appears to be a positive effect, suggesting that population increased by about 1.7%. This estimate is smaller than its counterpart for luminosity, suggesting a simultaneous increase in population (e.g. through migration) and a larger increase in wealth per capita. However, population data quality is poor for non-census years, so these results may be biased toward zero due to measurement error.

To strengthen the claim that changes in effort are the main driver of luminosity growth, analyzing the effects of reelection on public spending would also be useful. I leave this exercise for future work.

4.4 Wages and Treatment Effect

Theory also suggests that explicit monetary incentives and electoral career concerns may substitute for one another. If politicians already receive stronger monetary incentives (variable a in Section 2), the introduction of reelection may generate a smaller additional response. At the same time, supplemental payments (g), such as rents for long-term service might increase effects of reelections.

Mexico has a strict transparency policy which allows me to find data on salaries from Instituto Nacional de Transparencia, Acceso a la Información y Protección de Datos Personales (nd). In particular, for each municipality, I find the table of wages of government-employed workers (tabulador de sueldos y salarios), which typically contains the compensation of the municipal president. I calculate the fixed payment (monto neto), additions (percepciones adicionales, compensationes and dietas; e.g. compensacion especial por antiguedad), as well as the bonus payments (gratificaciones, primas, bonos, and estimulos).

As Instituto Nacional de Transparencia, Acceso a la Información y Protección de Datos Personales (nd) is a heavily protected governmental source and as the tables are not unified to a single format, it proves impossible to scrape this data, so I collect everything manually. Due to limitations in time, I only collect cross-sectional data for wages in the year 2025. It might prove interesting to extend the data collection and use time-variation in salaries for better identification strategies in further work.

I examine the theoretical predictions by relating the estimated municipality-month treatment effects from column (2) of Table **??** to local wage structure. The wage vari-

ables capture the level of compensation (fixed plus varying), the presence of any bonus payments, the share of bonuses in total wages, and the presence of supplements. Since wage structures are correlated with municipal size and development, the regressions also control for the baseline covariates used in Table 5.

Table 7: Wage Structure Heterogeneity

	Outcome Variable: Treatment Effect			
	(1)	(2)	(3)	(4)
Log Wages	−0.042	−0.043	−0.030	−0.029
	(0.007)	(0.007)	(0.007)	(0.006)
Any Bonus Payments		−0.046	−0.011	−0.012
		(0.007)	(0.006)	(0.007)
Share of Bonus in Wage		0.169	0.033	0.041
		(0.028)	(0.028)	(0.033)
Any Supplements		0.025	0.004	0.004
		(0.007)	(0.007)	(0.007)
Log Population	−0.060	−0.061	0.006	
	(0.004)	(0.004)	(0.004)	
Log GDP	−0.080	−0.078	0.012	0.012
	(0.004)	(0.004)	(0.004)	(0.004)
Baseline Luminosity			−0.104	−0.103
			(0.002)	(0.002)
Lambda				−0.001
				(0.006)
Observations	19379	19379	19379	385409

Columns 1–3 report OLS results, while column 4 reports Heckman results. The outcome variable is the treatment effect from column (2) in Table **??**. The regressors are described in more detail in the text. Controls are the regressors from column (1) in Table **??**. Standard errors are heteroskedasticity-robust.

Table 7 reports the results. The most robust pattern is that treatment effects are smaller in municipalities with higher wages. Across all columns, the coefficient on log wages is negative and precisely estimated, ranging from -0.043 to -0.029. This is consistent with the interpretation that the marginal value of reelection incentives is lower when politicians are already better compensated. In other words, higher baseline compensation may reduce the additional effort induced by the possibility of reelection.

The evidence on bonus payments is more mixed. In column (2), municipalities with any bonus payments have smaller treatment effects, while a larger bonus share is associated with larger treatment effects. However, both coefficients shrink substantially once baseline luminosity is included in columns (3) and (4). This suggests that the raw relationship between bonus structure and treatment-effect heterogeneity partly reflects broader differences across municipalities rather than the incentive scheme alone. Supplements are

also not consistently related to treatment effects after controls are included.

While formally law requires all municipalities to publish data on salaries, in the end, not all units comply. Sometimes this might indicate corruption, but in most cases these are very small municipalities that most likely do not pay any wages at all and hence have nothing to report. To address sample selection concerns in such a setup, I run the Heckman procedure in column 4 using population as the selection variable. The coefficients remain very similar to OLS results, and non-reporting municipalities appear to be negatively selected from the sample.

Note also that the negative coefficients on log population and log GDP in columns (1) and (2) disappear or change sign, while baseline luminosity is strongly negatively associated with the treatment effect. This suggests support for the mechanisms described in theory: the higher population is, the larger is the return to effort (variable b), calling for higher wages, and implying smaller implicit incentives (reelection effects). Population on its own is not a main driver of heterogeneity in treatment effects.

5 Conclusion

This paper studies how implicit electoral incentives interact with explicit monetary incentives for politicians. The theoretical model shows that the reelection concerns mechanism is not equally strong in all political environments. Career concerns are weaker when performance is noisy, when voters cannot easily infer effort, or when incumbents are electorally safe. In these cases, direct monetary incentives may become a useful complement to elections rather than a redundant or purely distortionary policy tool.

The empirical analysis tests the main prediction using Mexico's introduction of consecutive reelection for municipal presidents. Exploiting staggered implementation across states, I find that introducing reelection increased municipal nighttime luminosity. In the preferred staggered difference-in-differences specification, the effect is about 0.25 log points. The estimates remain positive across specifications with term-cycle controls, politician controls, municipality and time fixed effects, and municipality-specific trends and seasonality. Population-weighted estimates are smaller but still positive.

The pattern of heterogeneity in effects is mostly consistent with theory. The reform has positive effects throughout the term and especially in the middle of the political cycle. The effect is larger in smaller municipalities and weaker for politicians who won by larger margins, consistent with the idea that reelection incentives matter more when future office rents are valuable and when competition makes reelection responsive to effort. Notably, population effects disappear when including wages into heterogeneity estimations, supporting the mechanism of a switch to explicit incentives at a higher value of effort. Second-term politicians are less affected than first-term politicians, which is also consistent with weaker career concerns. Additional tests using winning margins, politician gender,

and population suggest that selection alone is unlikely to explain the results, although the available data do not allow a full separation of effort and selection mechanisms.

The main implication is that the design of wages for politicians should account for the strength of electoral accountability. Flat salaries may be sufficient when elections are competitive and informative, but they may be less appropriate when political competition is weak or when voters observe only noisy signals of performance. Conversely, performance pay may be most valuable precisely in settings where career concerns are limited. At the same time, in the setting of multitasking one should be cautious with simple performance schemes: monetary incentives should be tied to measures that are broad enough to avoid shifting effort only toward visible or easily measured tasks.

There are several limitations to resolve in further work. First, nighttime luminosity is an imperfect proxy for political effort and local performance. Using expenditure data for the same regressions could strenghten the claims. Second, politician-level data are limited, so the empirical analysis cannot fully distinguish changes in incentives (effort) from changes in selection (ability). Third, the wage component of the theoretical framework is not yet tested directly; doing so requires detailed information on local compensation rules and bonus payments. Future work should combine data on reelection incentives, compensation schemes, and public spending to estimate whether monetary incentives and career concerns are substitutes or complements in practice.

Overall, the results suggest that elections do discipline politicians, but only conditionally. Reelection incentives improve local outcomes when politicians have something to gain from future office and when political competition makes that future responsive to current performance. Understanding when these implicit incentives are strong, and when they need to be complemented by explicit compensation, is central for designing effective political institutions and should be addressed more carefully by policy setters.

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A Proofs and Derivations

This part of the appendix contains proofs and derivations left out of the main text. Subsection A.1 derives the variances of returns in the different setups of the model. Subsection A.2 considers the CARA case. Subsection A.3 examines the setup of multiple tasks. Finally, Subsection A.4 investigates the effects of distortions in political competition.

A.1 Variance of Returns and Other

This subsection addresses the variance of returns as well as related proofs and derivations.

A.1.1 Variance of Returns

To derive Equation 3, note that when η, θ, and ε are independent,

$$Var\left(f + a(t+\eta) - \frac{1}{2}ct^2 + \mathbb{I}\left[\theta + \varepsilon \geqslant \tilde{t}\right]\delta V\right) = Var(a\eta) + Var\left(\delta V \mathbb{I}\left[\theta + \varepsilon \geqslant \tilde{t}\right]\right)$$

The uncertainty of reelection can be calculated as

$$Var\left(\mathbb{I}(A)\right) = \int\int \left(\mathbb{I}(A) - \mathbb{P}(A)\right)^2 df_\theta df_\varepsilon = \int \mathbb{I}^2(A) - 2\mathbb{I}(A)\mathbb{P}(A) + \mathbb{P}^2(A) df_{\theta+\varepsilon} =$$

$$\int \mathbb{I}(A) df_{\theta+\varepsilon} - \mathbb{P}^2(A) = \mathbb{P}(A) - \mathbb{P}^2(A) = \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\left[1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right]$$

A.1.2 First-Order Effects of Effort on Variance

To show that reelection variance disappears in 5, note that

$$\frac{r\delta^2 V^2 \phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})}{2}\left[1 - 2\Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(\tilde{t})\right] = \frac{r\bar{\phi}\delta^2 V^2}{2}\left[1 - 2\cdot\frac{1}{2}\right] = 0$$

Intuitively, when ability is distributed symmetrically, effort only has second-order effects on the variance of reelection benefits in equilibrium, so the politician ignores them.

A.1.3 Variance of Returns with Endogenous Reelection Benefits

To derive equation 11, note that

$$\mathbb{E}\left[f + a(t+\eta) - \frac{1}{2}ct^2 + \delta\mathbb{I}(A)\left(g + b(t+\eta)\right)\right] = f + at - \frac{1}{2}ct^2 + \delta(g+bt)\mathbb{P}(A)$$

where A is true if $\theta + \varepsilon \geqslant \tilde{t}$. Also note that

$$Var\left(f + a(t+\eta) - \frac{1}{2}ct^2 + \delta\mathbb{I}(A)\left(g + b(t+\eta)\right)\right) =$$

$$\int\int\int \{a\eta + \delta\left[\mathbb{I}(A) - \mathbb{P}(A)\right]\left[g + bt\right] + \delta\mathbb{I}(A)b\eta\}^2 df_\theta df_\varepsilon df_\eta =$$

$$\int\int a^2\eta^2 + \delta^2\left[\mathbb{I}(A) - \mathbb{P}(A)\right]^2\left[g + bt\right]^2 + \delta^2\mathbb{I}(A)b^2\eta^2 +$$

$$2a\eta\delta\left[\mathbb{I}(A) - \mathbb{P}(A)\right]\left[g + bt\right] + 2\delta^2\mathbb{I}(A)b\eta\left[\mathbb{I}(A) - \mathbb{P}(A)\right]\left[g + bt\right] + 2\delta ab\eta^2\mathbb{I}(A)df_{\theta+\varepsilon}df_\eta$$

$$\int \left(a^2\eta^2 + \delta^2\left[\mathbb{P}(A) - \mathbb{P}^2(A)\right]\left[g + bt\right]^2 + \delta^2\mathbb{P}(A)b^2\eta^2 + 2\delta ab\eta^2\mathbb{P}(A)\right) df_\eta =$$

$$\delta^2\mathbb{P}(A)[1 - \mathbb{P}(A)](g + bt)^2 + (a + \delta b\mathbb{P}(A))^2\sigma_\eta^2 + \delta^2\mathbb{P}(A)[1 - \mathbb{P}(A)]b^2\sigma_\eta^2$$

A.1.4 Irrelevance of Future Bonus Payment b

Proposition 7. *It is never optimal to promise politicians bonuses paid for current effort in the future. Rather, paying unconditional bonuses in the present and promising a fixed payment in case of reelection is always more efficient.*

Suppose that the government can also pay bs in the future, in addition to R and g. The objective function of the politician in 1 now becomes

$$\max_{t\geqslant 0} f + at - \frac{1}{2}ct^2 + \delta(g + bt)\left[1 - \Phi\left(\tilde{t}\right)\right] - \tag{11}$$

$$\frac{1}{2}r\left\{\delta^2\left[1 - \Phi\left(\tilde{t}\right)\right]\Phi\left(\tilde{t}\right)(g + bt)^2 + \left(a + \delta b\left[1 - \Phi\left(\tilde{t}\right)\right]\right)^2\sigma_\eta^2 + \delta^2\left[1 - \Phi\left(\tilde{t}\right)\right]\Phi\left(\tilde{t}\right)b^2\sigma_\eta^2\right\}$$

To show that it is never optimal to set a positive b, let me derive the first-order condition:

$$\text{FOC:}\quad a - ct + \delta b\left[1 - \Phi\left(\tilde{t}\right)\right] + \delta(g + bt)\phi\left(\tilde{t}\right) -$$

$$r\left\{\delta^2 b\left[1 - \Phi\left(\tilde{t}\right)\right]\Phi\left(\tilde{t}\right)(g + bt) + \delta b\phi\left(\tilde{t}\right)\left(a + \delta b\left[1 - \Phi\left(\tilde{t}\right)\right]\right)\sigma_\eta^2\right\}$$

or

$$\left(1 - r\delta b\bar{\phi}\sigma_\eta^2\right)\left(a + \frac{\delta b}{2}\right) + \left(\delta\bar{\phi} - r\frac{\delta^2 b}{4}\right)(g + bt) = ct$$

Suppose in optimum $t = t^*$ and b is positive. Instead of a,g,b, take $b' = 0$ and

$$a' = \left(1 - r\delta b\bar{\phi}\sigma_\eta^2\right)\left(a + \frac{\delta b}{2}\right)$$

$$g' = \left(1 - r\frac{\delta b}{4\bar{\phi}}\right)(g + bt^*)$$

Note that t^* remains the same as the LHS of the FOC did not change. Now compare

$$t^* - f - at^* - \frac{\delta}{2}(g + bt^*) \quad \text{to} \quad t^* - f' - a't^* - \frac{\delta}{2}g'$$

where both f and f' satisfy IR. Then it is sufficient to compare variances

$$\frac{1}{2}r\left\{\frac{\delta^2(g+bt^*)^2}{4}+\left(a+\frac{\delta b}{2}\right)^2\sigma_\eta^2+\frac{\delta^2 b^2\sigma_\eta^2}{4}\right\}$$

and

$$\frac{1}{2}r\left\{\frac{\delta^2\left(1-r\frac{\delta b}{4\bar{\phi}}\right)^2(g+bt^*)^2}{4}+\left(1-r\delta b\bar{\phi}\sigma_\eta^2\right)^2\left(a+\frac{\delta b}{2}\right)^2\sigma_\eta^2\right\}$$

where the second expression is smaller. Hence, (f',a',g',0) achieves a strictly better result than (f,a,g,b), which means that it can never be optimal to set a positive b.

A.1.5 Voter Surplus

Note that the voter surplus can be calculated as

$$t^*-\frac{1}{2}c(t^*)^2-\frac{1}{2}r\left\{\frac{\delta^2(g^*)^2}{4}+(a^*)^2\sigma_\eta^2\right\}=$$

$$\frac{1}{c}\frac{1+4\bar{\phi}^2\sigma_\eta^2}{1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2}-\frac{1}{2c}\left(\frac{1+4\bar{\phi}^2\sigma_\eta^2}{1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2}\right)^2-\frac{cr\sigma_\eta^2}{2c}\left\{\frac{4\bar{\phi}^2\sigma_\eta^2+1}{(1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2)^2}\right\}=$$

$$\frac{1}{c}\frac{1+4\bar{\phi}^2\sigma_\eta^2}{1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2}-\frac{1}{2c}\frac{(1+4\bar{\phi}^2\sigma_\eta^2)(1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2)}{(1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2)^2}=\frac{1}{2c}\frac{1+4\bar{\phi}^2\sigma_\eta^2}{1+rc\sigma_\eta^2+4\bar{\phi}^2\sigma_\eta^2}$$

and a very similar derivation can be made for the case without reelection concerns.

A.2 The Case of CARA Utility

With a CARA function, expected utility becomes

$$-\exp\left\{-r\left(f+at-\frac{1}{2}ct^2\right)+\frac{1}{2}r^2a^2\sigma_\eta^2\right\}\left[\Phi(\tilde{t})+(1-\Phi(\tilde{t}))\exp\{-r\delta V\}\right]$$

or, after taking logarithms,

$$f+at-\frac{1}{2}ct^2-\frac{1}{2}r^2a^2\sigma_\eta^2-\frac{1}{r}\ln\left[\Phi(\tilde{t})+(1-\Phi(\tilde{t}))\exp\{-r\delta V\}\right]$$

Then the FOC implies

$$t=\frac{a+\frac{2\bar{\phi}}{r}\frac{1-\exp\{-r\delta V\}}{1+\exp\{-r\delta V\}}}{c}$$

so, very similar to 8,

$$a^*=\frac{1-\frac{2\bar{\phi}}{r}\frac{1-\exp\{-r\delta V\}}{1+\exp\{-r\delta V\}}}{1+rc\sigma_\eta^2}$$

which is qualitatively very similar to the result in the main text. The marginal value of reelection still depends on $\bar{\phi}$, so career concerns are weaker when voter inference is noisier. The only difference is that, under CARA utility, a^* remains positive even for large V.

A.3 The Case of Multiple Tasks

Suppose now that effort is a vector rather than a scalar, which is more realistic for political effort. Let $B(t) = t_1 + t_2 + \theta + \varepsilon$ and $C(t) = \frac{1}{2}c_1 t_1^2 + \frac{1}{2}c_2 t_2^2 + c_x t_1 t_2$. Also let the signal be $s = t + \eta$, with $\eta \sim N(0, \Sigma_\eta)$ and $\sigma_{12} = 0$. Defining $\tilde{t} = \bar{t} - t_1 - t_2$, 4 becomes

$$\max_{t \geqslant 0} \left\{ at - \frac{1}{2} t' C t + \delta V \left[1 - \Phi(\tilde{t})\right] - \frac{r\delta^2 V^2}{2} \left[\Phi(\tilde{t}) - \Phi^2(\tilde{t})\right] - \frac{1}{2} r a \Sigma_\eta a' \right\}$$

with the corresponding FOCs

$$a_i - c_i t_i - c_x t_{-i} + \delta \bar{\phi} V = 0$$

implying

$$t_1 = \frac{a_1 - c_x t_2 + \delta \bar{\phi} V}{c_1}$$

$$t_2 = \frac{a_2 - c_x t_1 + \delta \bar{\phi} V}{c_2}$$

or, after solving the system,

$$\left(1 - \frac{c_x^2}{c_1 c_2}\right) t_i = \frac{a_i - \frac{c_x}{c_{-i}} a_{-i} + \frac{c_{-i} - c_x}{c_{-i}} \delta \bar{\phi} V}{c_i}$$

or

$$t_i = \frac{c_{-i} a_i - c_x a_{-i} + (c_{-i} - c_x) \delta \bar{\phi} V}{c_1 c_2 - c_x^2}$$

Then the government objective function becomes

$$\frac{c_2 a_1 - c_x a_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2} + \frac{c_1 a_2 - c_x a_1 + (c_1 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2} -$$

$$-\frac{1}{2}c_1 \left(\frac{c_2 a_1 - c_x a_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2}\right)^2 - \frac{1}{2}c_2 \left(\frac{c_1 a_2 - c_x a_1 + (c_1 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2}\right)^2$$

$$-c_x \frac{c_2 a_1 - c_x a_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2} \cdot \frac{c_1 a_2 - c_x a_1 + (c_1 - c_x)\delta\bar{\phi}V}{c_1 c_2 - c_x^2} - \frac{1}{2} r \left\{ a_1^2 \sigma_{\eta_1}^2 + a_2^2 \sigma_{\eta_2}^2 + \frac{\delta^2 V^2}{4} \right\}$$

It is enough to derive the FOC for a_1, since the derivation for a_2 is symmetric:

$$\begin{aligned}
&\frac{c_2 - c_x}{c_1c_2 - c_x^2} - ra_1\sigma_{\eta_1}^2 - \frac{c_1c_2}{c_1c_2 - c_x^2}\frac{c_2a_1 - c_xa_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1c_2 - c_x^2} \\
&\quad + \frac{c_2c_x}{c_1c_2 - c_x^2}\frac{c_1a_2 - c_xa_1 + (c_1 - c_x)\delta\bar{\phi}V}{c_1c_2 - c_x^2} - \frac{c_xc_2}{c_1c_2 - c_x^2}\frac{c_1a_2 - c_xa_1 + (c_1 - c_x)\delta\bar{\phi}V}{c_1c_2 - c_x^2} \\
&\quad + \frac{c_x^2}{c_1c_2 - c_x^2}\frac{c_2a_1 - c_xa_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1c_2 - c_x^2} = 0
\end{aligned}$$

or

$$\frac{c_2 - c_x}{c_1c_2 - c_x^2} - ra_1\sigma_{\eta_1}^2 - \frac{c_2a_1 - c_xa_2 + (c_2 - c_x)\delta\bar{\phi}V}{c_1c_2 - c_x^2} = 0$$

implying

$$a_1 = \frac{(c_2 - c_x)(1 - \delta\bar{\phi}V) + c_xa_2}{c_2 + r(c_1c_2 - c_x^2)\sigma_{\eta_1}^2}$$

Then, if we do the same for a_2 and substitute it, we can get

$$a_1 = \frac{(c_2 - c_x)(1 - \delta\bar{\phi}V) + c_x\frac{(c_1 - c_x)(1 - \delta\bar{\phi}V) + c_xa_1}{c_1 + r(c_1c_2 - c_x^2)\sigma_{\eta_2}^2}}{c_2 + r(c_1c_2 - c_x^2)\sigma_{\eta_1}^2}$$

or

$$a_1 = \frac{(c_1c_2 - c_x^2)(1 - \delta\bar{\phi}V)(1 + r(c_2 - c_x)\sigma_{\eta_2}^2) + c_x^2a_1}{(c_1 + r(c_1c_2 - c_x^2)\sigma_{\eta_2}^2)(c_2 + r(c_1c_2 - c_x^2)\sigma_{\eta_1}^2)}$$

or

$$a_1 = \left(\frac{(1 + r(c_2 - c_x)\sigma_{\eta_2}^2)}{1 + c_2r\sigma_{\eta_2}^2 + c_1r\sigma_{\eta_1}^2 + r^2(c_1c_2 - c_x^2)\sigma_{\eta_1}^2\sigma_{\eta_2}^2}\right)(1 - \delta\bar{\phi}V)$$

General Multitasking

We can go one step further. Suppose effort has n dimensions and that $C(t)$ is some twice differentiable function. Suppose further that $B(t,\theta,\varepsilon) = b(t) + \theta + \varepsilon$ with $b(\cdot)$ being any differentiable function rather than $1't$. The proof below will not be new and will directly follow Holmström and Milgrom (1991) with the only addition of reelection concerns.

Note that first-order conditions imply

$$a_i + \delta\bar{\phi}V = C_i(t)$$

where subscripts on C denote partial derivatives. Then

$$\frac{\partial a}{\partial t} = [C_{ij}], \quad \text{so} \quad \frac{\partial t}{\partial a} = [C_{ij}]^{-1}$$

Then, one can derive the FOC of

$$B(t) - C(t) - \frac{1}{2} r a' \Sigma_\eta a$$

as

$$[C_{ij}]^{-1} B'(t) - [C_{ij}]^{-1}(a + (\delta \bar{\phi} V)_{n\times 1}) - r\Sigma_\eta a = 0$$

implying

$$a = (I + r[C_{ij}]\Sigma_\eta)^{-1} \left[B' - (\delta \bar{\phi} V)_{n\times 1}\right]$$

A.4 Distortions in Political Competition

Equation 5 simplifies because the equilibrium cutoff is evaluated at the point of maximal density, where reelection probability is one half. This relies on symmetry between the incumbent and potential challengers. I now relax this assumption by allowing challenger quality to be shifted downward. In particular, suppose the average ability of challengers is $\mathbb{E}[\theta_a] = -\alpha$, where $\alpha > 0$ measures the size of the distortion. This distortion can capture imperfect political competition, such as barriers to entry or repression of opponents, but it can also capture learning advantages enjoyed by incumbents.

With this distortion, the incumbent is more likely to be reelected at the same level of effort. The equilibrium reelection probability is $1 - \tilde{\Phi}_\alpha = 1 - \Phi_{\sigma_\theta^2+\sigma_\varepsilon^2}(-\alpha)$, and the relevant density at the cutoff is $\bar{\phi} \cdot \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2+\sigma_\varepsilon^2}\right\}$. Substituting these terms into 5 gives

$$ct = a + \delta \bar{\phi} V \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}\right\} + \frac{r\delta^2 V^2}{2} \bar{\phi} \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}\right\} [1 - 2\tilde{\Phi}_\alpha]$$

Following the same steps as in 6, the optimal explicit incentive is therefore

$$a^* = \max\left\{0, \frac{1 - \delta \bar{\phi} V \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2+\sigma_\varepsilon^2}\right\} - \frac{r\delta^2 V^2}{2} \bar{\phi} \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2+\sigma_\varepsilon^2}\right\} [1 - 2\tilde{\Phi}_\alpha]}{1 + rc\sigma_\eta^2}\right\}$$

This expression highlights two opposing mechanisms. First, a larger distortion α weakens electoral incentives because the equilibrium cutoff moves away from the point of maximal density. An additional unit of effort therefore has a smaller effect on reelection probability. Second, the variance term no longer drops out in equilibrium:

$$\frac{r\delta^2 V^2}{2} \bar{\phi} \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}\right\} [1 - 2\tilde{\Phi}_\alpha] > 0$$

This term means that effort also changes the risk associated with reelection benefits. As a result, the effect of political distortions on monetary incentives is not mechanical: distortions reduce the marginal electoral return to effort, but they also make the risk

channel relevant.

To determine which force dominates, differentiate the optimal bonus with respect to α:

$$\frac{\partial a}{\partial \alpha} = \frac{\alpha \delta V \bar{\phi} \exp\left\{-\frac{\alpha^2}{\sigma_\theta^2+\sigma_\varepsilon^2}\right\}\left[2 + r\delta V\left(1 - 2\tilde{\Phi}_\alpha - \bar{\phi}\exp\left\{-\frac{\alpha^2}{\sigma_\theta^2+\sigma_\varepsilon^2}\right\}\right)\right]}{1 + rc\sigma_{\eta_i}^2}$$

This derivative is positive as long as

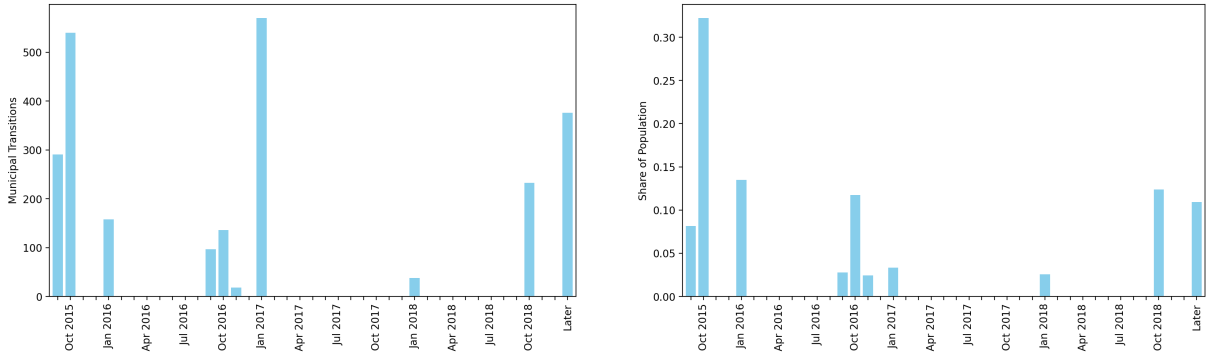
$$2 \geqslant \frac{r\delta V}{\sqrt{2\pi(\sigma_\theta^2 + \sigma_\varepsilon^2)}}$$

which is likely to hold when risk aversion is moderate and the marginal electoral value of effort is not too large. Under this condition, political distortions increase the optimal reliance on explicit compensation.

Proposition 8. *If the inequality above holds, then a larger distortion α lowers effort for any fixed bonus a. Hence, the optimal bonus must increase with α.*

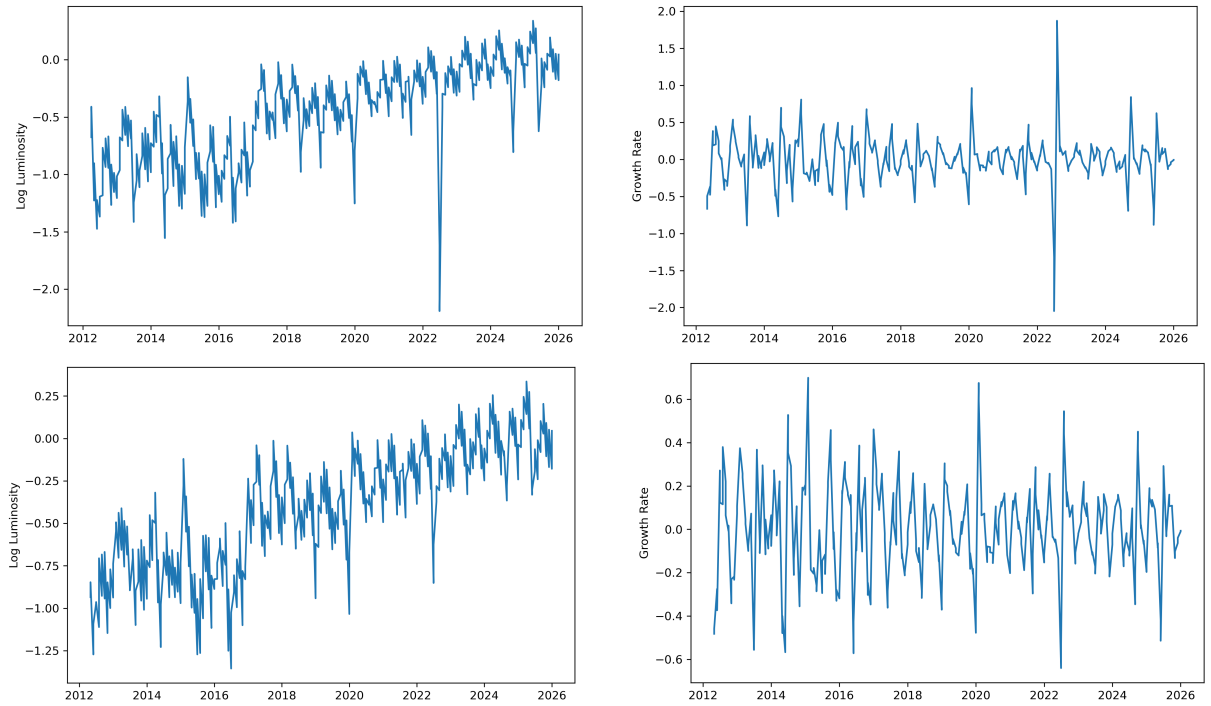
B Additional Figures

Figure 8: Transition Histograms



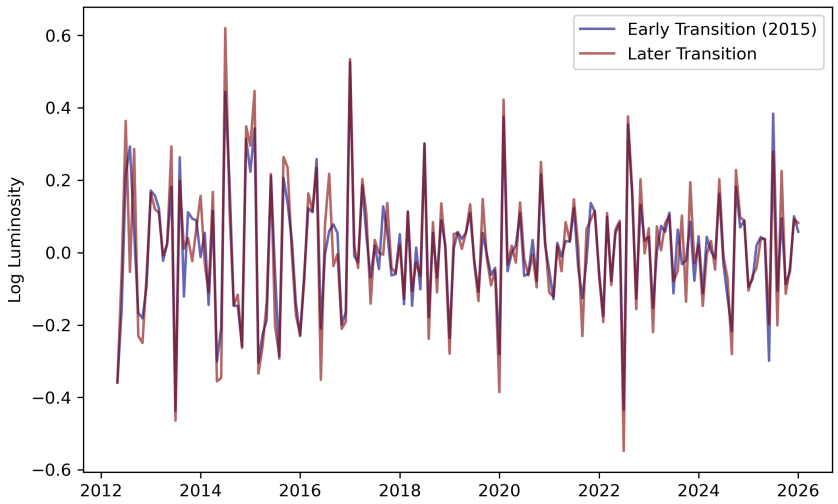
Note: This figure shows the distribution of municipal transitions over time (left) and a population-weighted analogue of that distribution.

Figure 9: Log Luminosity and Luminosity Growth



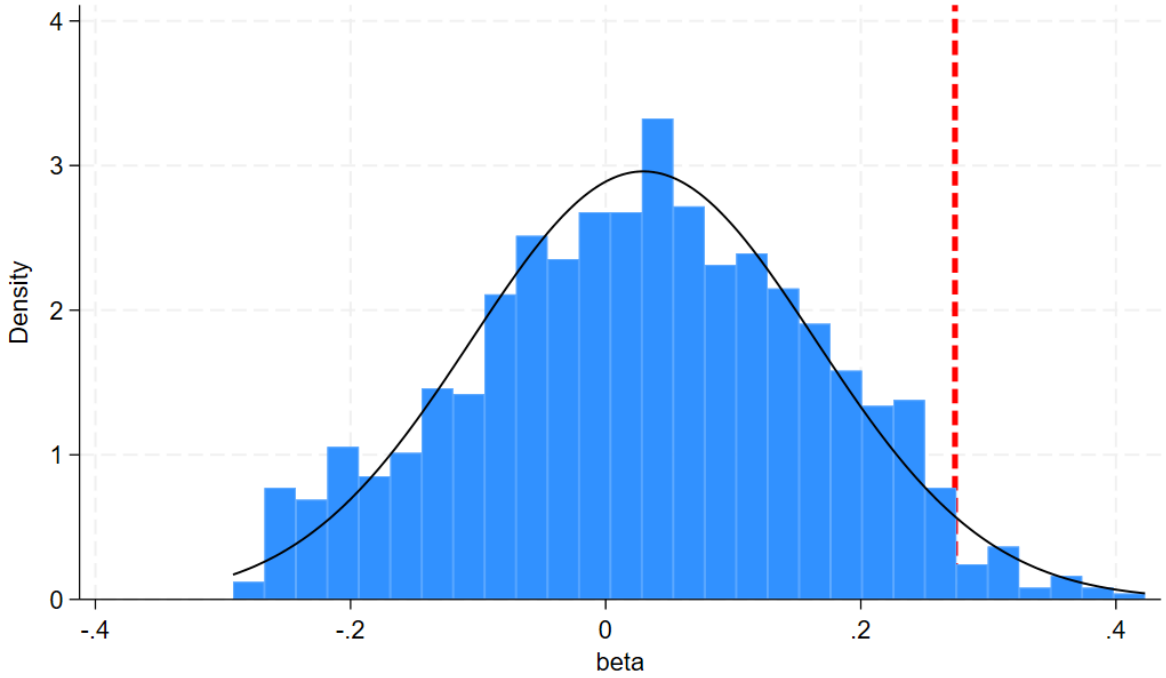
Note: This figure plots log luminosity (left) and its first difference (right) for the whole sample (top) and the truncated sample (bottom).

Figure 10: Log Luminosity Comparison



Note: This figure compares the main outcome variable, log mean nighttime luminosity in a given municipality-year, between groups with an early transition and a late transition. Data are taken from Elvidge et al. (2017).

Figure 11: Randomization Inference Results



Note: This figure shows the randomization inference results. Each estimation calculates the coefficient from column (2) of Table 3 with treatment randomly resampled at the state level. The p-value is 0.27.

C Additional Tables

Table 8: Non-Truncated Outcome (TWFE)

	Outcome Variable: Log Luminosity				
	(1) OLS	(2) OLS	(3) TWFE	(4) TWFE	(5) TWFE
Reelection Introduced	0.302	0.357	0.183	0.089	0.091
	(0.004)	(0.005)	(0.011)	(0.011)	(0.008)
Second Term				−0.003	0.006
				(0.006)	(0.004)
Term Length				0.044	−0.002
				(0.007)	(0.004)
Female				0.029	0.013
				(0.007)	(0.005)
Win Margin				0.116	−0.001
				(0.028)	(0.015)
Cycle Controls	No	Yes	Yes	Yes	Yes
Party FE	No	No	No	Yes	Yes
Municipality FE	No	No	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes
Mun Trend & Season	No	No	No	No	Yes
Errors	Robust	Robust	Mun Cl	Mun Cl	Mun Cl
Observations	403,111	403,111	403,111	280,155	280,155
Mean of Dep. Variable	−0.485	−0.485	−0.485	−0.381	−0.381
Adj. R-squared	0.012	0.031	0.902	0.915	0.931

This table reports the same calculations as Table 2 in the main text, except that the outcome variable is not truncated.

Table 9: Unregular States Dropped (TWFE)

	Outcome Variable: Log Luminosity				
	(1) OLS	(2) OLS	(3) TWFE	(4) TWFE	(5) TWFE
Reelection Introduced	0.456	0.506	0.093	0.071	0.073
	(0.005)	(0.005)	(0.008)	(0.010)	(0.007)
Second Term				0.002	0.006
				(0.006)	(0.003)
Term Length				0.032	−0.006
				(0.006)	(0.003)
Female				0.024	0.010
				(0.007)	(0.004)
Win Margin				0.093	0.001
				(0.024)	(0.012)
Cycle Controls	No	Yes	Yes	Yes	Yes
Party FE	No	No	No	Yes	Yes
Municipality FE	No	No	Yes	Yes	Yes
Time FE	No	No	Yes	Yes	Yes
Mun Trend & Season	No	No	No	No	Yes
Errors	Robust	Robust	Mun Cl	Mun Cl	Mun Cl
Observations	345,140	345,140	345,140	263,102	263,102
Mean of Dep. Variable	−0.485	−0.485	−0.485	−0.381	−0.381
Adj. R-squared	0.031	0.049	0.930	0.936	0.953

This table reports the same calculations as Table 2 in the main text, except that the states of Veracruz, Hidalgo, and Ciudad de Mexico are dropped from the sample

Table 10: Non-Truncated Outcome (DiD)

	Outcome Variable: Log Luminosity			
	(1) SDiD	(2) SDiD	(3) SDiD	(4) P-W SDiD
Reelection Introduced	0.308	0.265	0.357	0.147
	(0.016)	(0.022)	(0.041)	(0.058)
Term Controls	Yes	Yes	Yes	Yes
Politician Controls	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Mun Trend & Season	No	No	Yes	Yes
Observations	403,111	250,173	247,270	242,856
Mean of Dep. Variable	−0.485	−0.425	−0.427	−0.451

This table reports the same calculations as Table 3 in the main text, except that the outcome variable is not truncated.

Table 11: Unregular States Dropped (DiD)

	Outcome Variable: Log Luminosity			
	(1) SDiD	(2) SDiD	(3) SDiD	(4) P-W SDiD
Reelection Introduced	0.305	0.228	0.278	0.092
	(0.021)	(0.028)	(0.037)	(0.043)
Term Controls	Yes	Yes	Yes	Yes
Politician Controls	No	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Mun Trend & Season	No	No	Yes	Yes
Observations	235,655	179,994	179,654	176,245
Mean of Dep. Variable	−0.485	−0.425	−0.427	−0.451

This table reports the same calculations as Table 3 in the main text, except that the states of Veracruz, Hidalgo, and Ciudad de Mexico are dropped from the sample

Table 12: DiD Multiple GT

	Outcome Variable: Log Luminosity			
	(1) DiD-MGT	(2) DiD-MGT	(3) DiD-MGT	(4) DiD-MGT
Weighted Average Slope	0.154	0.164	0.105	0.126
	(0.013)	(0.016)	(0.014)	(0.009)
Controls	Yes	Yes	Yes	Yes
Mun & Time FE	Yes	Yes	Yes	Yes
Trunctuated	Yes	No	Yes	Yes
Dropped States	No	No	Yes	No
State Trend & Season	No	No	No	Yes
Observations	20,979	21,763	17,076	20,979
Mean of Dep. Variable	−0.425	−0.425	−0.425	−0.425

The estimates report DiD-calculations as in De Chaisemartin and d'Haultfoeuille (2024). The controls are the same as in Table 3 column (2).

Table 13: Treatment Effect Heterogeneity

	Outcome Variable: Log Luminosity			
	(1) Second Term	(2) High Win Margin	(3) Female	(4) Cartel Presence
Yes	0.297	0.284	0.324	0.142
	(0.029)	(0.024)	(0.031)	(0.028)
No	0.255	0.247	0.240	0.285
	(0.021)	(0.022)	(0.021)	(0.022)
Observations	252,563	252,563	252,563	252,468

The regression is the same as column (2) in Table 3 with the treatment effect heterogenized by politicians in their first vs second terms, those with high (above median) vs low winning margins and females vs males as well as by municipalities with cartel presence vs without.